THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM THREE ANNUAL EXAMINATION
031 PHYSICS

Time: 3 Hours **November 2025**

Instructions

1. This paper consists of Sections **A**, **B**, and **C** with a total of eleven questions.
2. Answer all questions in Sections **A** and **B**, and only two questions from Section **C**.
3. Unauthorized materials and devices are not allowed in the examination room.
4. Non-programmable calculators and mathematical tables may be used.
5. Write your Examination Number on every page.
6. Use these constants where necessary:
7. $g = 10\ m/s^2$
8. Density of mercury = $13{,}600\ kg/m^3$
9. Density of air = $1.25\ kg/m^3$
10. $\pi = 3.14$
11. Speed of sound = 340 m/s

SECTION A (16 MARKS)
Answer all questions in this section

1. For each item (i) – (x), choose the correct answer from among the given alternatives and write its letter besides the item number in the answer booklet provided

   (i) A student sees her whole face in a mirror fixed on a wall. Which type of mirror is she using?
   A. Concave mirror
   B. Plane mirror
   C. Convex mirror
   D. Parabolic mirror
   E. Cylindrical mirror

   (ii) A body continues to move in a straight line at constant speed unless acted upon by another force. This statement describes:
   A. Newton's second law
   B. Newton's first law
   C. Newton's third law
   D. The law of moments
   E. The law of gravitation

   (iii) When a plastic ruler is rubbed with dry hair, it becomes charged because:
   A. It gains protons

B. It loses protons
C. It gains electrons
D. It loses electrons
E. Electrons are created

(iv) The part of a bar magnet where the magnetic force is strongest is called:
A. Magnetic axis
B. Magnetic field
C. Magnetic pole
D. Neutral point
E. Equatorial line

(v) When a car suddenly stops, passengers move forward because of:
A. Centripetal force
B. Inertia of motion
C. Friction
D. Gravity
E. Pressure

(vi) Which of the following is not an example of work being done?
A. A student lifting a bag
B. A ball rolling on a flat table
C. A man pushing a cart
D. A woman climbing stairs
E. A porter carrying a load upstairs

(vii) Which of the following is a non-renewable energy source?
A. Sunlight
B. Wind
C. Coal
D. Water
E. Biomass

(viii) The correct unit for pressure in the SI system is:
A. Pascal
B. Watt
C. Newton
D. Joule
E. Kilogram

(ix) The device used to measure very small thickness, such as that of paper, is:
A. Micrometer screw gauge
B. Vernier calipers
C. Ruler
D. Tape measure
E. Compass

(x) When a solid changes directly into gas without becoming liquid, the process is called:

A. Fusion
B. Condensation
C. Evaporation
D. Sublimation
E. Freezing

2. Match the descriptions of the terms used in pressure in List A with its corresponding concept used in pressure in List B by writing the letter of the correct response beside the corresponding item number in the answer booklet provided

List A	List B
(i) Hydraulic jack	A. Used to measure the gas pressure
(ii) Pascal's principle	B. Used to measure atmospheric pressure
(iii)Barometer	C. Liquid used in thermometers and barometers
(iv)Manometer	D. Upward force on a floating body
(v)) Buoyant force	E. Pressure is transmitted equally in fluids
(vi) Mercury	F. Application of Pascal's principle
	G. Low pressure
	H. High pressure

SECTION B (54 MARKS)
Answer all questions in this section

3. (a) Two students at Seypalm secondary school observe themselves using mirrors. Student A uses a plane mirror, and Student B uses a convex mirror. Compare the images they see in terms of size and nature.
(b) A student places a candle in front of a convex lens and sees a diminished image on a screen. State the position of the candle and give a reason for your answer.

4. (a) State two laws of vector addition.
(b) Two forces of 25 N and 35 N act on a body at an angle of 90°. Calculate the resultant force and the angle it makes with the 25 N force.

5. (a) A wooden box weighing 500 N is pulled up a smooth plank 3 m long to reach a height of 0.6 m using a force of 120 N. Find the efficiency of the plane.
(b) A 0.15 kg ball moving at 18 m/s hits a wall and rebounds at 12 m/s in the opposite direction in 0.1 s. Find the average force on the ball.

6. (a) At sea level, the pressure is 76 cm of mercury. At the top of a hill, it is 70 cm. Estimate the height of the hill.
(b) A uniform beam of weight 300 N is 4 m long and supported at both ends. If a 200N load is placed 1 m from one end, calculate the reactions at the supports.

7. (a) A student drops a lump of clay onto the floor, and it sticks. What type of collision is this, and how are momentum and kinetic energy affected?

(b) A ball dropped from a height of 16 m rebounds to 9 m. Find the coefficient of restitution.

8. (a) State four uses of solar energy in daily life.
   (b) Draw a simple diagram showing how different energy sources depend on the Sun.

SECTION C (30 MARKS)
Answer any two questions from this section

9. (a) Three resistors of 4 Ω, 6 Ω, and 12 Ω are connected in parallel.
   (i) Draw their circuit diagram.
   (ii) Calculate the total resistance.
   (b) Three capacitors of 5 μF, 10 μF, and 15 μF are connected in parallel. Find the total capacitance and state one advantage of connecting capacitors this way.
   (c) (i) Define lightning.
   (ii) State four ways of protecting buildings from lightning damage.

10. (a) Explain the difference between additive and subtractive color mixing using examples from real life.
    (b) Mention two devices that use the principle of total internal reflection.
    (c) A 2 cm tall object is placed 12 cm in front of a concave mirror of focal length 8 cm. Find the image position, magnification, and nature of the image formed.

11. (a) Two bicycles of the same type carry different loads - one heavy, one light. After a 10 km journey, the heavily loaded bicycle's tyres are more worn out. Explain this observation.
    (b) Why is the coefficient of friction a number without units?
    (c) A block of 3 kg slides down an incline at 30° with an acceleration of 4 m/s^2. Calculate the coefficient of friction.
    (d)Suppose there was no friction on the Earth's surface:
    (i) List three activities that would be impossible.
    (ii) Explain two benefits of friction.